

bayesstuff

Sviatoslav

2026-09-01

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.2.1    ✓ readr      2.2.0
## ✓ forcats   1.0.1    ✓ stringr   1.6.0
## ✓ ggplot2    4.0.3    ✓ tibble    3.3.1
## ✓ lubridate 1.9.5    ✓ tidyr     1.3.2
## ✓ purrr     1.2.2
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
gmppanel <- read_csv("~/ (No subject)/Courseworkstuff/New folder (2)/gmppanel.txt")
```

```
## Rows: 168 Columns: 4
## — Column specification —
## Delimiter: ","
## chr (2): Location, Classification
## dbl (2): Year, Count
##
## ⓘ Use `spec()` to retrieve the full column specification for this data.
## ⓘ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
gmppanel_prop1 <- gmppanel %>%
  group_by(Location, Year) %>%
  mutate(Proportion = Count / sum(Count, na.rm = TRUE)) %>%
  ungroup()
gmppanel_prop2 <- read_csv("C:/Users/alexa/Downloads/gmppanel_prop.csv")
```

```
## Rows: 264 Columns: 6
## — Column specification —
## Delimiter: ","
## chr (3): Location, Area, Classification
## dbl (3): Year, Count, Proportion
##
## ⓘ Use `spec()` to retrieve the full column specification for this data.
## ⓘ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
gmppanel_prop2=gmppanel_prop2[,-2]
gmppanel_prop=full_join(gmppanel_prop1,gmppanel_prop2)
```

```
## Joining with `by = join_by(Location, Year, Classification, Count, Proportion)`
```

```

high_peak_areas <- c(
  "New Mills West Ward",
  "New Mills East Ward",
  "Buxton Central Ward",
  "Dinting Ward",
  "Old Glossop Ward",
  "Hadfield South Ward",
  "Hadfield North Ward"
)

unemployed <- na.omit(filter(gmppanel_prop,Classification=="Never worked & long-term unemployed"))
unemployed <- unemployed %>%
  filter(!(Location %in% c("Tameside District","High Peak District")),Year>=1971) %>%
  mutate(treatment = factor(ifelse(Location %in%high_peak_areas, 1, 0)),
         established = ifelse(Year>=2011,1,0),
         did= established*treatment)

```

```

## Warning: There was 1 warning in `mutate()`.
## i In argument: `did = established * treatment`.
## Caused by warning in `Ops.factor()`:
## ! '*' not meaningful for factors

```

```

library(rstanarm)

```

```

## Loading required package: Rcpp
## This is rstanarm version 2.32.2
## - See https://mc-stan.org/rstanarm/articles/priors for changes to default priors!
## - Default priors may change, so it's safest to specify priors, even if equivalent to the defaults.
## - For execution on a local, multicore CPU with excess RAM we recommend calling
##   options(mc.cores = parallel::detectCores())

```

```

library(marginaleffects)
library(bayestestR)
set.seed(123)
spline_random <- stan_glmer(
  Proportion ~ factor(treatment)*factor(Year)+(1 | Location),
  data = unemployed,
  family = gaussian(),cores=8,chains=8
)
grid = unemployed[,c(2,7)]
grid$Year = as.factor(grid$Year)
paths = predictions(spline_random,ndraws=4000,re.form=NA)

```

```

## Warning: These arguments are not known to be supported for models of class
## `stanreg`: ndraws, re.form. All arguments are still passed to the
## model-specific prediction function, but users are encouraged to check if the
## argument is indeed supported by their modeling package. Please file a request
## on Github if you believe that an unknown argument should be added to the
## `marginaleffects` white list of known arguments, in order to avoid raising this
## warning: https://github.com/vincentarelbundock/marginaleffects/issues

```

```

pathpred = paths[,c(2,3,4,6,10)]
pathpred = data.frame(pathpred)
ggplot(pathpred, aes(x = Year, y = estimate, color = treatment, group = treatment)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  # Add Bayesian uncertainty ribbons
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, fill = treatment),
    alpha = 0.15, color = NA) + scale_color_manual(
    name = "Boundary Side",
    values = c("orange", "darkgreen"),
    labels = c("Greater Manchester", "High Peak") # <--- UPDATE THIS
  ) +
  scale_fill_manual(
    name = "Boundary Side",
    values = c("orange", "darkgreen"),
    labels = c("Greater Manchester", "High Peak") # <--- MUST MATCH EXACTLY
  ) + ggtitle(unique(unemployed$Classification)) + labs(subtitle = "95 percent probability that the true v
alue lies in shaded area(posterior interval)")

```

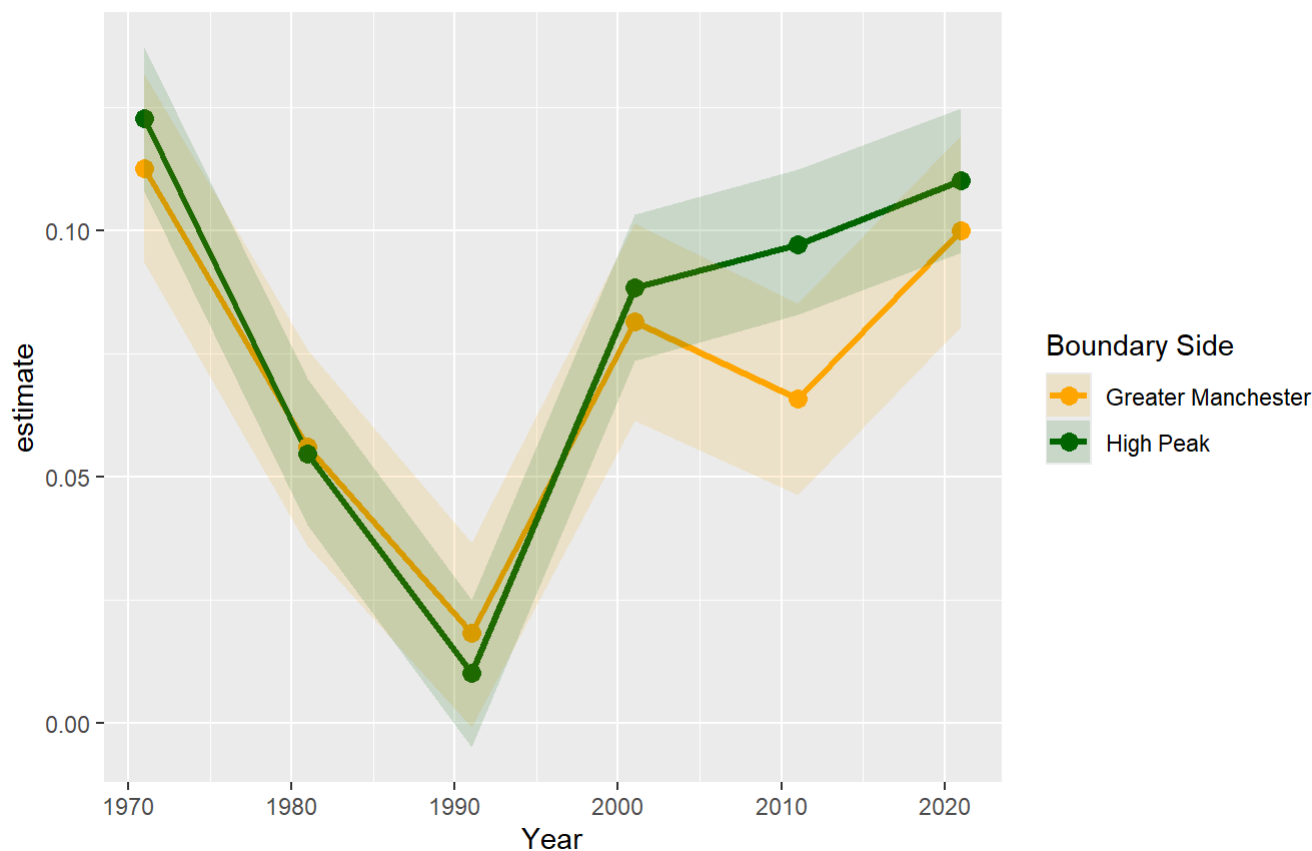
```

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

Never worked & long-term unemployed

95 percent probability that the true value lies in shaded area(posterior interval)



```
describe_posterior(spline_random)
```

```
## Summary of Posterior Distribution
##
## Parameter | Median | 95% CI | pd
## -----|-----|-----|-----
## (Intercept) | 0.11 | [ 0.09, 0.13] | 100%
## factor(treatment)1 | 9.97e-03 | [-0.01, 0.03] | 79.33%
## factor(Year)1981 | -0.06 | [-0.08, -0.04] | 100%
## factor(Year)1991 | -0.09 | [-0.11, -0.07] | 100%
## factor(Year)2001 | -0.03 | [-0.05, -0.01] | 99.66%
## factor(Year)2011 | -0.05 | [-0.07, -0.03] | 100%
## factor(Year)2021 | -0.01 | [-0.03, 0.01] | 87.89%
## factor(treatment)1:factor(Year)1981 | -0.01 | [-0.04, 0.01] | 80.49%
## factor(treatment)1:factor(Year)1991 | -0.02 | [-0.04, 0.01] | 91.84%
## factor(treatment)1:factor(Year)2001 | -3.39e-03 | [-0.03, 0.02] | 59.90%
## factor(treatment)1:factor(Year)2011 | 0.02 | [-0.01, 0.05] | 94.65%
## factor(treatment)1:factor(Year)2021 | 2.62e-04 | [-0.03, 0.03] | 50.85%
##
## Parameter | ROPE | % in ROPE | Rhat | ESS (tail)
## -----|-----|-----|-----|-----
## (Intercept) | [ 0.00, 0.00] | 0% | 1.001 | 4726
## factor(treatment)1 | [ 0.00, 0.00] | 20.07% | 1.000 | 4132
## factor(Year)1981 | [ 0.00, 0.00] | 0% | 1.002 | 4991
## factor(Year)1991 | [ 0.00, 0.00] | 0% | 1.001 | 4835
## factor(Year)2001 | [ 0.00, 0.00] | 0% | 1.001 | 5227
## factor(Year)2011 | [ 0.00, 0.00] | 0% | 1.001 | 5130
## factor(Year)2021 | [ 0.00, 0.00] | 15.53% | 1.001 | 4829
## factor(treatment)1:factor(Year)1981 | [ 0.00, 0.00] | 17.89% | 1.002 | 5236
## factor(treatment)1:factor(Year)1991 | [ 0.00, 0.00] | 9.42% | 1.001 | 4895
## factor(treatment)1:factor(Year)2001 | [ 0.00, 0.00] | 23.95% | 1.001 | 4907
## factor(treatment)1:factor(Year)2011 | [ 0.00, 0.00] | 6.64% | 1.001 | 4679
## factor(treatment)1:factor(Year)2021 | [ 0.00, 0.00] | 24.97% | 1.001 | 4564
```

```
gmppanel <- read_csv("~/ (No subject)/Courseworkstuff/New folder (2)/gmppanel.txt")
```

```
## Rows: 168 Columns: 4
## — Column specification —————
## Delimiter: ","
## chr (2): Location, Classification
## dbl (2): Year, Count
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
gmppanel_prop1 <- gmppanel %>%
  group_by(Location, Year) %>%
  mutate(Proportion = Count / sum(Count, na.rm = TRUE)) %>%
  ungroup()
gmppanel_prop2 <- read_csv("C:/Users/alexa/Downloads/gmppanel_prop.csv")
```

```
## Rows: 264 Columns: 6
## — Column specification —————
## Delimiter: ","
## chr (3): Location, Area, Classification
## dbl (3): Year, Count, Proportion
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
gmppanel_prop2=gmppanel_prop2[,-2]
gmppanel_prop=full_join(gmppanel_prop1,gmppanel_prop2)
```

```
## Joining with `by = join_by(Location, Year, Classification, Count, Proportion)`
```

```
high_peak_areas <- c(
  "New Mills West Ward",
  "New Mills East Ward",
  "Buxton Central Ward",
  "Dinting Ward",
  "Old Glossop Ward",
  "Hadfield South Ward",
  "Hadfield North Ward"
)
unemployed <- na.omit(filter(gmppanel_prop,Classification=="Intermediate occupations"))
unemployed <- unemployed %>%
  filter(!(Location %in% c("Tameside District","High Peak District")),Year>=1971) %>%
  mutate(treatment = factor(ifelse(Location %in%high_peak_areas, 1, 0)),
         established = ifelse(Year>=2011,1,0),
         did= established*treatment)
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `did = established * treatment`.
## Caused by warning in `Ops.factor()`:
## ! '*' not meaningful for factors
```

```
library(rstanarm)
library(marginaleffects)
library(bayestestR)
set.seed(123)
spline_random <- stan_glmer(
  Proportion ~ factor(treatment)*factor(Year)+(1 | Location),
  data = unemployed,
  family = gaussian(),cores=8,chains=8
)
grid = unemployed[,c(2,7)]
grid$Year = as.factor(grid$Year)
paths = predictions(spline_random,ndraws=4000,re.form=NA)
```

```
## Warning: These arguments are not known to be supported for models of class
## `stanreg`: ndraws, re.form. All arguments are still passed to the
## model-specific prediction function, but users are encouraged to check if the
## argument is indeed supported by their modeling package. Please file a request
## on Github if you believe that an unknown argument should be added to the
## `marginaleffects` white list of known arguments, in order to avoid raising this
## warning: https://github.com/vincentarelbundock/marginaleffects/issues
```

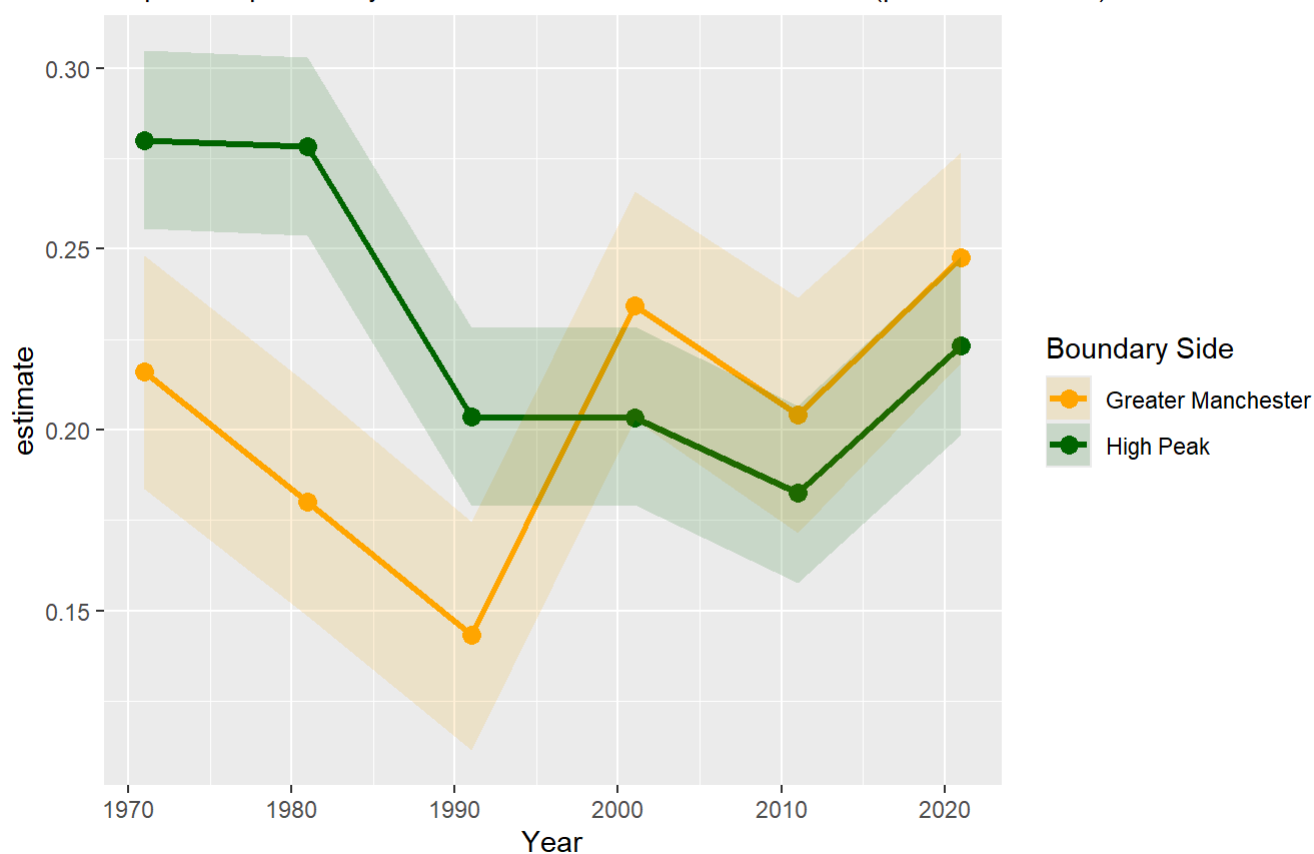
```

pathpred = paths[,c(2,3,4,6,10)]
pathpred = data.frame(pathpred)
ggplot(pathpred, aes(x = Year, y = estimate, color = treatment, group = treatment)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  # Add Bayesian uncertainty ribbons
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, fill = treatment),
            alpha = 0.15, color = NA) + scale_color_manual(
            name = "Boundary Side",
            values = c("orange","darkgreen"),
            labels = c("Greater Manchester","High Peak") # <--- UPDATE THIS
          ) +
  scale_fill_manual(
    name = "Boundary Side",
    values = c("orange","darkgreen"),
    labels = c("Greater Manchester","High Peak") # <--- MUST MATCH EXACTLY
  ) + ggtitle(unique(unemployed$Classification))+labs(subtitle = "95 percent probability that the true v
alue lies in shaded area(posterior interval)")

```

Intermediate occupations

95 percent probability that the true value lies in shaded area(posterior interval)



```
describe_posterior(spline_random)
```

Summary of Posterior Distribution

##	Parameter	Median	95% CI	pd	
##	-----	-----	-----	-----	
##	(Intercept)	0.22	[0.18, 0.25]	100%	
##	factor(treatment)1	0.06	[0.02, 0.10]	99.86%	
##	factor(Year)1981	-0.04	[-0.08, 0.01]	94.81%	
##	factor(Year)1991	-0.07	[-0.12, -0.03]	99.95%	
##	factor(Year)2001	0.02	[-0.02, 0.06]	80.30%	
##	factor(Year)2011	-0.01	[-0.06, 0.03]	71.11%	
##	factor(Year)2021	0.03	[-0.01, 0.07]	93.25%	
##	factor(treatment)1:factor(Year)1981	0.03	[-0.02, 0.09]	89.08%	
##	factor(treatment)1:factor(Year)1991	-3.43e-03	[-0.06, 0.05]	54.57%	
##	factor(treatment)1:factor(Year)2001	-0.09	[-0.15, -0.04]	99.98%	
##	factor(treatment)1:factor(Year)2011	-0.09	[-0.14, -0.03]	99.81%	
##	factor(treatment)1:factor(Year)2021	-0.09	[-0.14, -0.03]	99.98%	
##					
##	Parameter	ROPE	% in ROPE	Rhat	ESS (tail)
##	-----	-----	-----	-----	-----
##	(Intercept)	[0.00, 0.00]	0%	1.001	4646
##	factor(treatment)1	[0.00, 0.00]	0%	1.003	4495
##	factor(Year)1981	[0.00, 0.00]	4.75%	1.001	5331
##	factor(Year)1991	[0.00, 0.00]	0%	1.001	5484
##	factor(Year)2001	[0.00, 0.00]	12.80%	1.001	5510
##	factor(Year)2011	[0.00, 0.00]	15.75%	1.001	5377
##	factor(Year)2021	[0.00, 0.00]	6.84%	1.001	5442
##	factor(treatment)1:factor(Year)1981	[0.00, 0.00]	7.08%	1.002	5604
##	factor(treatment)1:factor(Year)1991	[0.00, 0.00]	14.34%	1.001	5175
##	factor(treatment)1:factor(Year)2001	[0.00, 0.00]	0%	1.001	5409
##	factor(treatment)1:factor(Year)2011	[0.00, 0.00]	0%	1.002	4895
##	factor(treatment)1:factor(Year)2021	[0.00, 0.00]	0%	1.001	5303

```

unemployed <- na.omit(filter(gmppanel_prop,Classification=="Higher managerial, administrative & profess
ional occupations" ))
unemployed <- unemployed %>%
  filter(Location != "Tameside District",Year>=1971) %>%
  mutate(treatment = factor(ifelse(Location == "High Peak District", 1, 0)),
         established = ifelse(Year>=2011,1,0),
         did= established*treatment)

```

Warning: There was 1 warning in `mutate()`.
i In argument: `did = established \* treatment`.
Caused by warning in `Ops.factor()`:
! '*' not meaningful for factors

```

set.seed(123)
spline_random <- stan_glmer(
  Proportion ~ factor(treatment)*factor(Year)+(1 | Location),
  data = unemployed,
  family = gaussian(),cores=8,chains=8
)
grid = unemployed[,c(2,7)]
grid$Year = as.factor(grid$Year)
paths = predictions(spline_random,ndraws=4000,re.form=NA)

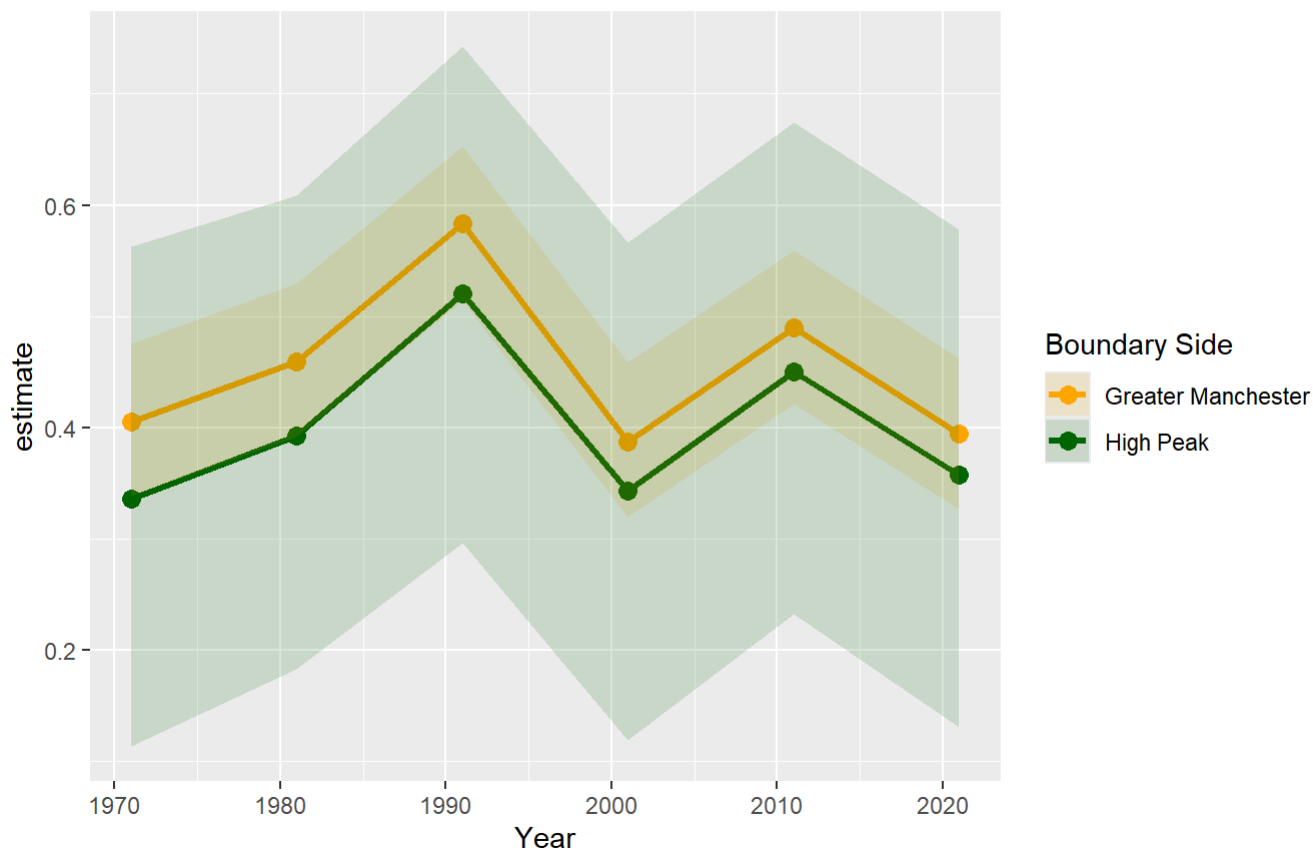
```

```
## Warning: These arguments are not known to be supported for models of class
## `stanreg`: ndraws, re.form. All arguments are still passed to the
## model-specific prediction function, but users are encouraged to check if the
## argument is indeed supported by their modeling package. Please file a request
## on Github if you believe that an unknown argument should be added to the
## `marginaleffects` white list of known arguments, in order to avoid raising this
## warning: https://github.com/vincentarelbundock/marginaleffects/issues
```

```
pathpred = paths[,c(2,3,4,6,10)]
pathpred = data.frame(pathpred)
ggplot(pathpred, aes(x = Year, y = estimate, color = treatment, group = treatment)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  # Add Bayesian uncertainty ribbons
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, fill = treatment),
    alpha = 0.15, color = NA) + scale_color_manual(
    name = "Boundary Side",
    values = c("orange", "darkgreen"),
    labels = c("Greater Manchester", "High Peak") # <--- UPDATE THIS
  ) +
  scale_fill_manual(
    name = "Boundary Side",
    values = c("orange", "darkgreen"),
    labels = c("Greater Manchester", "High Peak") # <--- MUST MATCH EXACTLY
  ) + ggtitle(unique(unemployed$Classification)) + labs(subtitle = "95 percent probability that the true v
alue lies in shaded area(posterior interval)")
```

Higher managerial, administrative & professional occupations

95 percent probability that the true value lies in shaded area(posterior interval)



```
describe_posterior(spline_random)
```

```
## Summary of Posterior Distribution
##
## Parameter | Median | 95% CI | pd
## -----|-----|-----|-----
## (Intercept) | 0.41 | [ 0.34, 0.48] | 100%
## factor(treatment)1 | -0.07 | [-0.31, 0.16] | 73.25%
## factor(Year)1981 | 0.05 | [ 0.01, 0.10] | 99.56%
## factor(Year)1991 | 0.18 | [ 0.14, 0.22] | 100%
## factor(Year)2001 | -0.02 | [-0.06, 0.02] | 80.75%
## factor(Year)2011 | 0.08 | [ 0.05, 0.13] | 100%
## factor(Year)2021 | -0.01 | [-0.05, 0.03] | 71.71%
## factor(treatment)1:factor(Year)1981 | 4.42e-03 | [-0.13, 0.13] | 52.49%
## factor(treatment)1:factor(Year)1991 | 4.99e-03 | [-0.14, 0.15] | 53.06%
## factor(treatment)1:factor(Year)2001 | 0.03 | [-0.12, 0.16] | 63.90%
## factor(treatment)1:factor(Year)2011 | 0.03 | [-0.11, 0.17] | 65.81%
## factor(treatment)1:factor(Year)2021 | 0.03 | [-0.11, 0.17] | 67.53%
##
## Parameter | ROPE | % in ROPE | Rhat | ESS (tail)
## -----|-----|-----|-----|-----
## (Intercept) | [-0.01, 0.01] | 0% | 1.005 | 2745
## factor(treatment)1 | [-0.01, 0.01] | 6.64% | 1.001 | 3892
## factor(Year)1981 | [-0.01, 0.01] | 0% | 1.000 | 4812
## factor(Year)1991 | [-0.01, 0.01] | 0% | 1.000 | 4805
## factor(Year)2001 | [-0.01, 0.01] | 33.63% | 1.001 | 5528
## factor(Year)2011 | [-0.01, 0.01] | 0% | 1.000 | 4853
## factor(Year)2021 | [-0.01, 0.01] | 41.64% | 1.001 | 5224
## factor(treatment)1:factor(Year)1981 | [-0.01, 0.01] | 16.04% | 1.002 | 4725
## factor(treatment)1:factor(Year)1991 | [-0.01, 0.01] | 14.67% | 1.001 | 5223
## factor(treatment)1:factor(Year)2001 | [-0.01, 0.01] | 13.03% | 1.000 | 5104
## factor(treatment)1:factor(Year)2011 | [-0.01, 0.01] | 13.47% | 1.001 | 4794
## factor(treatment)1:factor(Year)2021 | [-0.01, 0.01] | 12.93% | 1.001 | 5142
```

```
unemployed <- na.omit(filter(gmppanel_prop,Classification=="Routine & manual occupations" ))
unemployed <- unemployed %>%
  filter(Location != "Tameside District",Year>=1971) %>%
  mutate(treatment = factor(ifelse(Location == "High Peak District", 1, 0)),
         established = ifelse(Year>=2011,1,0),
         did= established*treatment)
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `did = established * treatment`.
## Caused by warning in `Ops.factor()`:
## ! '*' not meaningful for factors
```

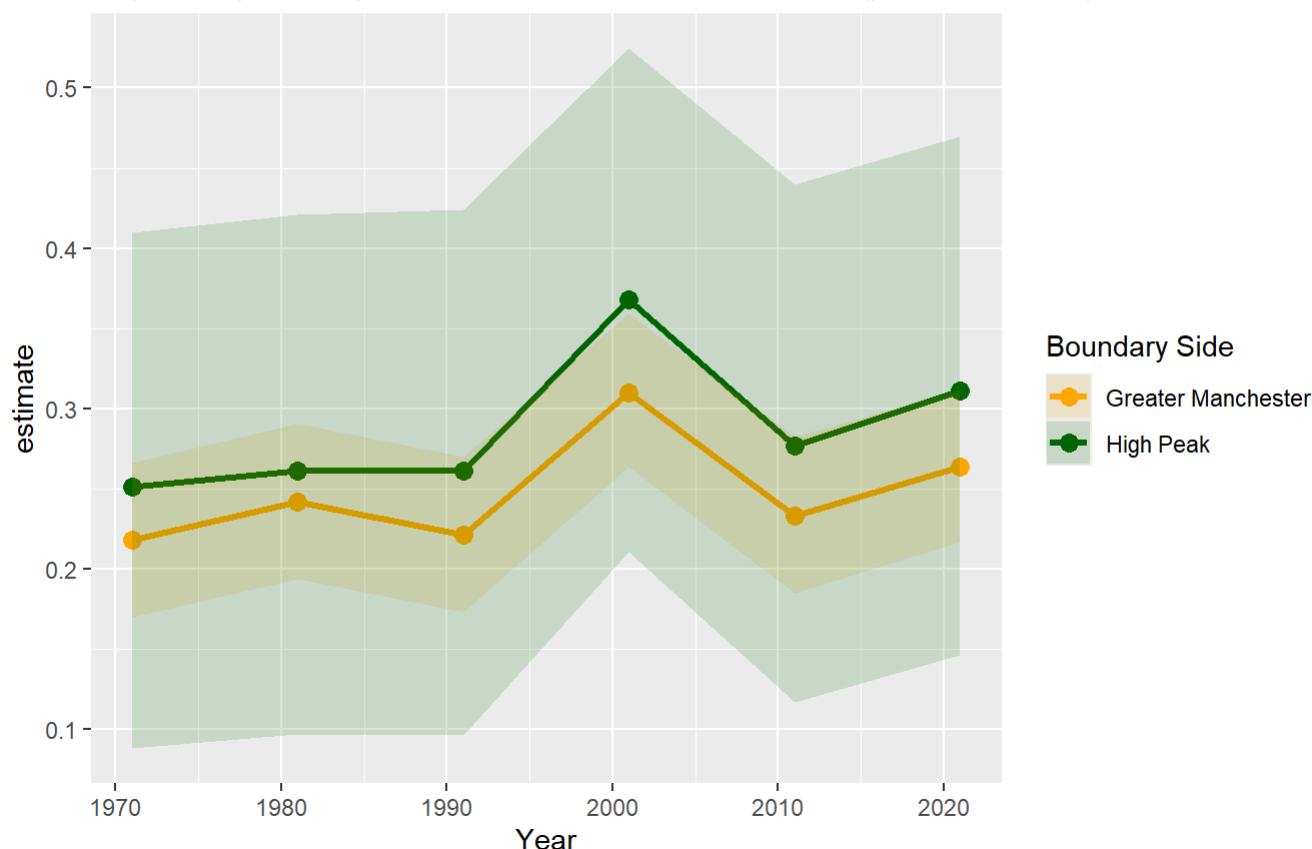
```
set.seed(123)
spline_random <- stan_glmer(
  Proportion ~ factor(treatment)*factor(Year)+(1 | Location),
  data = unemployed,
  family = gaussian(),cores=8,chains=8
)
grid = unemployed[,c(2,7)]
grid$Year = as.factor(grid$Year)
paths = predictions(spline_random,ndraws=4000,re.form=NA)
```

```
## Warning: These arguments are not known to be supported for models of class
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## warning: https://github.com/vincentarelbundock/marginaleffects/issues
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```
pathpred = paths[,c(2,3,4,6,10)]
pathpred = data.frame(pathpred)
ggplot(pathpred, aes(x = Year, y = estimate, color = treatment, group = treatment)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  # Add Bayesian uncertainty ribbons
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, fill = treatment),
    alpha = 0.15, color = NA) + scale_color_manual(
    name = "Boundary Side",
    values = c("orange", "darkgreen"),
    labels = c("Greater Manchester", "High Peak") # <--- UPDATE THIS
  ) +
  scale_fill_manual(
    name = "Boundary Side",
    values = c("orange", "darkgreen"),
    labels = c("Greater Manchester", "High Peak") # <--- MUST MATCH EXACTLY
  ) + ggtitle(unique(unemployed$Classification)) + labs(subtitle = "95 percent probability that the true v
alue lies in shaded area(posterior interval)")
```

Routine & manual occupations

95 percent probability that the true value lies in shaded area(posterior interval)



```
describe_posterior(spline_random)
```

Summary of Posterior Distribution

##

## Parameter	Median	95% CI	pd
## -----			
## (Intercept)	0.22	[0.17, 0.27]	100%
## factor(treatment)1	0.03	[-0.14, 0.20]	65.99%
## factor(Year)1981	0.02	[0.00, 0.05]	97.00%
## factor(Year)1991	2.93e-03	[-0.02, 0.03]	59.62%
## factor(Year)2001	0.09	[0.07, 0.12]	100%
## factor(Year)2011	0.01	[-0.01, 0.04]	87.36%
## factor(Year)2021	0.05	[0.02, 0.07]	99.96%
## factor(treatment)1:factor(Year)1981	-0.01	[-0.10, 0.07]	62.74%
## factor(treatment)1:factor(Year)1991	5.43e-03	[-0.08, 0.09]	54.83%
## factor(treatment)1:factor(Year)2001	0.02	[-0.05, 0.10]	73.89%
## factor(treatment)1:factor(Year)2011	0.01	[-0.08, 0.10]	59.62%
## factor(treatment)1:factor(Year)2021	0.01	[-0.07, 0.10]	62.20%

##

## Parameter	ROPE	% in ROPE	Rhat	ESS (tail)
## -----				
## (Intercept)	[-0.01, 0.01]	0%	1.003	2852
## factor(treatment)1	[-0.01, 0.01]	7.96%	1.001	4750
## factor(Year)1981	[-0.01, 0.01]	8.96%	1.000	5848
## factor(Year)1991	[-0.01, 0.01]	51.46%	1.000	5895
## factor(Year)2001	[-0.01, 0.01]	0%	1.000	5651
## factor(Year)2011	[-0.01, 0.01]	28.47%	1.000	5684
## factor(Year)2021	[-0.01, 0.01]	0%	1.000	5573
## factor(treatment)1:factor(Year)1981	[-0.01, 0.01]	15.39%	1.001	4667
## factor(treatment)1:factor(Year)1991	[-0.01, 0.01]	16.17%	1.001	5189
## factor(treatment)1:factor(Year)2001	[-0.01, 0.01]	15.76%	1.001	5191
## factor(treatment)1:factor(Year)2011	[-0.01, 0.01]	16.04%	1.001	5265
## factor(treatment)1:factor(Year)2021	[-0.01, 0.01]	15.34%	1.001	5264